01 / 50 W functional design

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Design result: a stationary, battery-free 12 V DC supply for a 50 W load, with a calculated 16-hour no-solar interval after charging. This document specifies a functional electrical design and prototype acceptance criteria. No assembled hardware, measured performance or fabrication-qualified PCB is claimed.

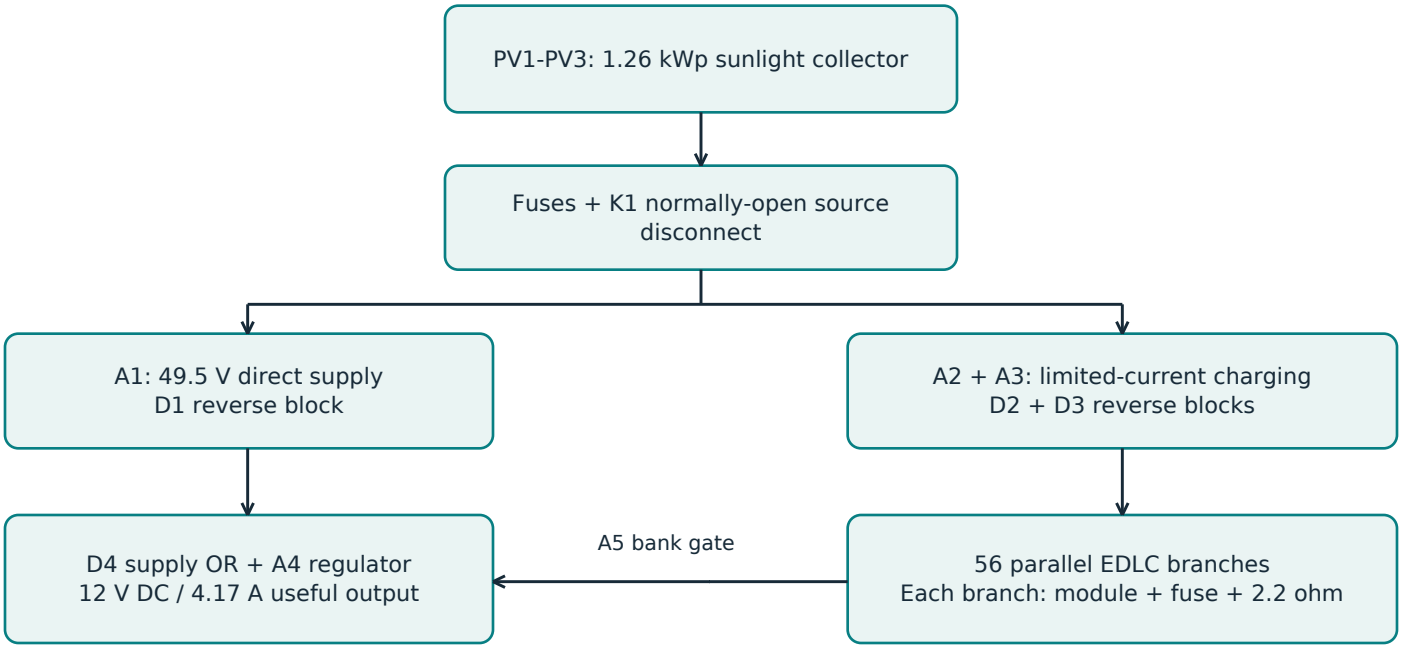
Requirement	Selected design
Useful service	12.0 V DC nominal, 4.17 A, 50 W; output acceptance 11.4-12.6 V at a constant 50 W load.
Environmental energy	Sunlight only during operation. No grid supply, fuel, electrochemical battery or self-powered internal loop.
Collector	Three REC420AA PURE-R panels in parallel, 1,260 Wp total. [S1]
Storage	56 Eaton XLR-48R6167-R EDLC modules in parallel through individual protected, resistive branches. [S2-S3]
Continuity envelope	At least 45.5 V at each storage module before a no-solar interval of at most 16 hours; then a qualified replenishment window.
Daily replenishment	Eight-hour source window delivering at least 288 W combined to the direct and storage paths, after their conversion/interconnect losses, while service continues.
Size consequence	823.2 kg of capacitor modules alone [S16], approximately 5.80 m2 of panel surface. Racks, mounting, wiring and electronics add size and mass.

This changes the original RF/static/coil harvesting mechanism. Sunlight is the ambient resource selected here. The design does not establish 50 W from ordinary indoor RF, static charge or incidental vibration. A 50 W round-the-clock load uses 1.2 kWh/day at the output.

What is finished, and what requires a prototype

Finished here: power topology, compatible voltage ranges, operating setpoints, board-change schedule, branch arrangement, energy sizing, control behavior and commissioning sequence. Remaining: detailed control-board layout, fuse/resistor pulse coordination, loop-stability and transient verification, and physical service testing. These are explicit engineering release gates, not completed results.

02 / Electrical topology and wiring



All negative returns join at B0. PV auxiliary power starts before K1 closes.

Connection	Implementation
PV bus	PV1-PV3 each through a 15 A DC fuse into a common combiner. Use parallel connections only; do not series-connect the three panels.
K1 input/output	Combined PV positive enters K1; K1 output feeds separate A1/A2/A3 input fuses. PV negative goes to B0. A PV-only auxiliary supply is connected before K1.
Direct path	A1 output positive -> D1 anode; D1 cathode -> SUPPLY+. A1 negative -> B0. This is the preferred path whenever its voltage exceeds the bank path.
Charge paths	A2/A3 output positive -> their own D2/D3 anodes; both cathodes -> CB+. Separate diodes prevent a charged bank feeding either converter backward.
Storage branches	For each M1-M56: positive terminal -> local 2 A fuse -> 2.2 ohm resistor -> CB+; negative terminal -> B0. Never use a module terminal as the common distribution point.
Bank load path	CB+ -> 5 A load fuse -> A5 bank hot-swap gate -> D4 anode; D4 cathode -> SUPPLY+.
Useful output	SUPPLY+ and B0 -> A4 input. A4 isolated 12 V output -> 5 A output fuse -> load connector. Define a single intentional output-return bond if required by the connected load.

D1-D4: STPS10H100CFP, one package per path; join both anodes on each device and use its common cathode. Mount each on its own suitable heatsink. The diode orientation is part of the design: reversing one would defeat the intended power path. [S7]

03 / Source and conversion circuits

A1-A3 use ADI DC1924A assemblies based on LT8705. This controller uses voltage and current loops rather than a chemistry-specific battery-start algorithm. Retain the vendor power stage, compensation, current shunt and switching layout; make only the listed setting changes for the first prototype. [S4-S5]

Setting	A1 direct supply	A2/A3 chargers
Output target, before diode	49.5 V	46.5 V
Output current limit, nominal	1.99 A	4.96 A each
R49, IMON_OUT resistor	86.6 kohm	34.8 kohm
R13, output-divider top	392 kohm + 20 kohm sealed trim	374 kohm + 5 kohm sealed trim
R16, output-divider bottom	10.0 kohm, 0.1%	10.0 kohm, 0.1%
R53/R54, input regulation	590 kohm / 20.0 kohm	649 kohm / 20.0 kohm
Input-regulation target	36.75 V nominal	40.31 V nominal
R55, default FBIN link	Remove before fitting R53/R54	Remove before fitting R53/R54
R11/R14, input UVLO	590 kohm / 20.0 kohm	590 kohm / 20.0 kohm
Mode jumper	Position marked DCM	Position marked DCM

R13/R16, R53/R54 and R11/R14 are named as on the published DC1924A schematic. Do not apply these designators to another revision without checking its schematic. The final R13 value is set by measurement; nominal resistor arithmetic is not calibration.

Why this starts and shares power

A1 can supply the load without waiting for the large bank to charge. A2/A3 charge a discharged bank under current control; there is no battery-detection or chemical charge timer in this chosen topology. Their higher input-voltage regulation point makes charging yield before the direct path when PV voltage falls. A1 is not wired to charge the bank. This is fixed input-voltage regulation, not a claim of true MPPT.

The three panels have a calculated cold open-circuit bound of 70.73 V at -40 C including the stated +3% voltage tolerance; parallel wiring retains that voltage. It is below the DC1924A 80 V input ceiling. A1/A2/A3 are used only in the qualified input range; panel wattage is not treated as guaranteed delivered power. [S1, S4]

Prototype release requires: A1 stays 48.4-50.5 V; A2/A3 stay 46.3-46.8 V before their diodes; each charger supplies at least 4.5 A when current-limited under the source envelope, and never exceeds 5.5 A. Verify transitions from zero bank voltage, including controller temperature and output-loop stability.

04 / Storage and fault-current design

Item	Design value
Module / count	Eaton XLR-48R6167-R / 56 identical modules. No module series connection.
Capacitance used in sizing	132.8 F per module, 80% of 166 F; bank effective C = 7,436.8 F. This is a required acceptance minimum.
Voltage window used in sizing	45.5 V minimum initial module voltage down to a conservative 22.0 V endpoint.
Module leakage allowance	No more than 10 mA per complete module over the qualified operating range, after conditioning. Aggregate allowance 0.56 A.
ESR / branch resistor	Accept module ESR no greater than 10 milliohm. Add TE HSC2002R2J, 2.2 ohm / 200 W / 5%, in each positive branch. [S10]
Branch fuse	Littelfuse KLKD 2 A, DC-rated, fitted at the positive terminal before the resistor. [S9]
Normal branch current	Around 0.18 A average with 10 A aggregate charging. Do not assume equal sharing until checked.
Equalization before connection	All module voltages within 0.1 V; connect only while the common bus is isolated from charging and loads.

The resistors are intentional

An unfettered parallel EDLC bank has extreme prospective fault current. With every branch passing through at least 2.09 ohm (2.2 ohm minus 5%), and module voltage at or below 48 V, the resistive upper bound is 22.97 A per branch, or 1,286 A for 56 branches. Include up to 27 A from the PV source and local filter-capacitor transients in the fault study. Wiring inductance and fuse action are not credited in that resistive bound.

At night the branch network adds only about 0.1-0.4 W at normal total current; the power budget reserves 0.5 W. During charging it dissipates several watts across the whole bank. Each resistor must survive the measured clearing-energy pulse from its branch fuse. Its 200 W nameplate alone does not establish pulse survival.

Release gate: the selected branch fuse must clear the defined branch fault before the resistor, terminal, insulation or wire energy limit is exceeded. Confirm time-current and pulse-energy coordination with the component data and a bounded test fixture. Do not perform a direct short on the assembled bank. A fault between a module terminal and its local fuse is outside the resistor-limited segment and requires insulated, protected construction.

Use a common CB+ busbar and B0 return with separate, equal-length module branches. Add a 15 A bank-trunk fuse and a DC-rated manual isolation device. An isolation switch does not discharge the modules. Follow Eaton handling, mounting and discharge instructions. [S3]

05 / Protection and control behavior

A5 is an LM5069-1 bank-load gate, implemented from the TI reference circuit with an appropriately rated external pass MOSFET. This is a custom circuit adaptation, not an unmodified evaluation-board claim. The bank gate is used on the small load path, never as a timed inrush switch for charging the entire bank. [S6]

Circuit / setting	Implementation
Undervoltage divider	Use independent UVLO/OVLO divider arrangement. UVLO top 523 kohm, bottom 70.6 kohm, both 0.1%. Nominal disconnect 21.02 V, reconnect 32.00 V.
Gate current limit	15 milliohm, 1%, Kelvin sense resistor: nominal 3.67 A for a 55 mV threshold. Accept 3.3-4.1 A, with no loss of 50 W service at minimum bank voltage.
Fault timing / pass stage	1.0 uF timer (about 47 ms nominal); 31.6 kohm power-limit resistor (about 20 W at 48 V). Pass-device reference: CSD19536KTT, 100 V; mount for a case temperature below 60 C and confirm derated SOA. [S15]
Independent charge overvoltage	TLV1702 + LM4040A25: compare CB+ divided by 178 kohm / 10.0 kohm, 0.1%, with 2.500 V. Nominal trip 47.0 V; qualify 46.7-47.3 V. [S11-S12]
Hardwired fault action	A storage overvoltage, module alarm, thermal-loop opening or emergency stop sets a fault latch; open K1, force A2/A3 SHDN low and force A5 UVLO low.
Module alarms	Connect each Eaton J1 alarm to a 5 V-or-lower pull-up and wired alarm collection, within the vendor sink-current limit. Add separate continuity supervision; an open-collector alarm by itself cannot prove its cable is connected. [S3]
Thermal behavior	Qualified storage ambient 10-35 C. Fit normally-closed, calibrated 45 C body-temperature switches in a supervised loop; include converter heatsink protection. Use individual internal thermistor readings during commissioning.
Reset	Manual reset only after a latched fault has cleared. Undervoltage is a normal source-exhaustion state; A5 may reconnect at its rising threshold after charging.

Power and the cold-start interlock

Provide a PV-only 12 V auxiliary supply before K1, using a TEN 60-7212WIR; it powers the K1 coil and a 5 V regulator. A separate RECOM R-78HB5.0-0.5 from the bank supplies night controls. Diode-OR the 5 V rails. Night auxiliary input must remain at or below 3 W. Budget 10 W PV-only auxiliary demand during the qualified daytime window; check contactor pickup separately. [S8, S13-S14]

K1 reference: TE EV200AAANA, normally open. Its coil driver must default off when control power is lost. Use a fault-dominant latch, power-on inhibit and healthy-loop interlock; no logging processor may override the hardware fault line. A4 enable = healthy AND (A1 output-good OR A5 power-good); hold A4 disabled during bank-gate startup. Qualify no more than 330 uF switched by A5 and confirm startup finishes before the minimum fault timeout. Bond the PV auxiliary output return to B0 for the controls. The control-board layout, temperature-switch model, continuity circuitry and SOA verification remain prototype engineering details requiring review.

06 / Energy budget with explicit losses

Quantity	Calculation	Result
Useful output	50 W x 24 h	1,200 Wh/day
Worst output-path draw	50 W / 0.85	58.824 W
Night controls	Accepted maximum	3.000 W
Conservative leakage bound	56 x 0.010 A x 47.5 V	26.600 W
Branch resistors / ESR allowance	Accepted maximum at load	0.500 W
Worst combined bank draw	58.824 + 3 + 26.6 + 0.5	88.924 W
Required for 16 hours	88.924 W x 16 h	1,422.776 Wh
Usable stored energy	0.5 x 7,436.8 F x (45.5^2 - 22^2) / 3,600	1,638.420 Wh
Energy remaining beyond 16 h	1,638.420 - 1,422.776	215.644 Wh
Calculated energy-bound autonomy	1,638.420 / 88.924	18.425 h

The 85% value is an acceptance minimum for the complete useful-output path, including A5, D4, wiring and A4 conversion. It is not a measured efficiency. The 0.5 W storage-branch allowance is additional, so it is not hidden in that efficiency. The leakage term deliberately uses a higher voltage than the initial energy calculation. All auxiliary inputs, displays, monitoring, heaters or fans must fit the stated allowances or force resizing.

Daily restoration without discharging the bank over successive days

For a deliberately conservative daily screen, apply the same 88.924 W bound for all 24 hours: 2,134.165 Wh/day. An eight-hour accepted delivery window of 288 W supplies 2,304 Wh, leaving 169.835 Wh/day for restoration margin or curtailment. This is a power-and-energy requirement at the post-conversion boundary, not a forecast from panel nameplates.

As an input-side screen, 340 W of actual usable PV input, minus 10 W PV auxiliaries, at 90% conversion/interconnection efficiency yields 297 W, above the 288 W requirement. Qualify that input under the real PV I-V curve and controller current limits. A daily energy total without the required timing cannot establish continuity.

What the calculation establishes: the selected storage and operating envelope have enough calculated energy for the declared service. It does not establish that a specific assembly meets the capacitance, leakage, efficiency, temperature or source criteria. Those bounds are measured acceptance requirements.

07 / Daily operation and source loss

State	What the circuit does	Service boundary
Initial / discharged	PV auxiliary powers controls before K1. After healthy-loop check and arming, A1 starts from PV; A2/A3 charge the bank in current limit.	50 W is available only when PV can support the direct path. No overnight promise from an empty bank.
Charging with service	A1 feeds SUPPLY+ through D1. Chargers feed CB+ through D2/D3. D4 prevents direct-path voltage charging the bank through its load port.	Qualify post-conversion delivery at 288 W or more over the eight-hour window.
Ready for night	Confirm all module voltages at least 45.5 V, measured parameters within bounds and no active faults.	The 16-hour no-solar interval starts from this qualified state.
Solar declines	Charging yields at the higher input-regulation setting. As the direct path drops below bank voltage, D4 conducts.	A4 regulates 12 V throughout the admitted input range. Handover transients must meet output acceptance.
Night	Bank supplies the useful load, control electronics and losses. Solar converters cannot pull energy backward through D2/D3.	50 W for 16 hours is the calculated design requirement, with the stated initial conditions.
Prolonged source loss	A5 disconnects at undervoltage. Storage then supplies only residual standby leakage/controls until their own UVLO.	The device cannot promise continuous output through unlimited darkness.
Fault	Latch fault; open K1, inhibit charge controllers and open bank load gate.	Explicit repair/check and manual reset precede service.

Do not confuse charging time with output startup

The direct path avoids making the useful load wait while several kilowatt-hours of gross capacitor energy are accumulated. Charging the bank from zero may span more than one solar day. The first full-capacity night begins only after the ready condition is achieved; there is no arbitrary five-second or single-day full-bank startup claim.

Resource qualification

Evaluate the proposed site using a measured or independently modeled PV time series and the actual fixed-voltage control behavior. Clouds, shade, soiling, temperature and seasons can prevent the replenishment window. A shorter input window, lower delivered power or longer source interruption requires more collector capacity, more storage or a second environmental source. No weather record or site qualification was supplied in this task.

08 / Component and assembly schedule

Qty	Reference / selection	Purpose
3	REC420AA PURE-R	Parallel solar collector, 1.26 kWp total.
3	ADI DC1924A	A1 direct supply; A2/A3 charging. Apply page 3 settings.
56	Eaton XLR-48R6167-R	Electrostatic storage; acceptance limits on pages 4 and 6.
56	TE HSC2002R2J	2.2 ohm branch current-limiting resistors, mounted on metal supports.
56	Littelfuse KLKD 2 A + rated holders	One positive-terminal fuse per storage module.
3/1	KLKD 15 A / KLKD 30 A	PV branch fuses / main source fuse; coordinate with the panel, wiring and disconnect.
3	KLKD 5 A, 10 A, 10 A	A1/A2/A3 input-branch fuses respectively.
1 ea	KLKD 15 A, 5 A, 5 A	Bank trunk, bank load branch and 12 V output protection.
4	STPS10H100CFP + heatsinks	D1-D4 reverse blocks / supply OR. Join the two anodes per package.
1	Traco TEN 60-4812WIR	A4, 18-75 V input, regulated 12 V / 5 A output.
1	Traco TEN 60-7212WIR	PV-only 12 V auxiliary supply. Confirm pickup-current capability for K1.
2	RECOM R-78HB5.0-0.5	5 V rails from PV auxiliary and bank; diode-OR their outputs.
1	TE EV200AAANA + rated driver	K1 source disconnect; coil power from PV auxiliary only.
1	LM5069-1 circuit adaptation	A5 load-path UVLO, inrush and current protection; pages 5 and 10.
set	TLV1702, LM4040A25, latch/interlock	Independent charge cutoff and fault-dominant control.
56+	NC temperature switches + harness	Calibrated thermal loop, continuity supervision and module alarm collection.
4	Anchored racks, at least 300 kg each	14 modules per rack with insulated terminal covers.
set	DC isolators, busbars, guards, cables	PV conductors rated for the real installation; 100 V-or-higher circuit accessories.

This is a functional component schedule, not a supplier quote or a fabrication BOM for the unfinished control PCB. It does not invent stock status, purchase prices or a qualified substitute. Use official board schematics for unchanged board components. The selected pass MOSFET still needs SOA verification. Thermal switches and final fuse/resistor coordination are not procurement-release selections. Auxiliary feeds also need dedicated DC fuses selected for their measured pickup and operating currents.

09 / Physical arrangement and commissioning

Mechanical and thermal arrangement

Use four stationary racks with fourteen modules per rack, placed horizontally and individually secured. Provide terminal covers, separate protected busbars and strain relief. The 823.2 kg storage mass excludes frames and branch resistors. Select anchorage and supporting floor from actual loads; this system is not a desktop or portable power box. Place the electronics in a separate shaded enclosure with natural or budgeted forced cooling.

Use common return B0 and star distribution. Initial reference sizing: 6 mm2 copper for the bank trunk, 2.5 mm2 for short protected module branches and the 12 V output, and PV cable appropriate to the combiner current, route and temperature. These are short-run reference sizes, not a site ampacity or voltage-drop approval. Limit the 12 V output loop resistance to 0.05 ohm, including connectors; verify load-terminal voltage at 4.17 A. Use a locking connector rated at least 10 A DC.

Assembly and release sequence

Step	Required action and evidence
1. Inspect / characterize	Keep modules isolated. Follow vendor handling and terminal instructions. Characterize capacitance, ESR, complete-module leakage and alarm behavior; condition and measure at the qualified temperatures. Reject or resize if bounds fail.
2. Set converters on a bounded supply	Before connecting the large bank, calibrate A1-A3 with a current-limited bench source and small test capacitor. Test input priority, zero-voltage charging, current limits, reverse blocks and 12 V regulation. Bench power is test energy, not harvested output.
3. Validate protection separately	Inject voltage/alarm/thermal signals. Verify K1 opens and A5 disconnects with power loss and faults. Validate MOSFET SOA, fuse pulse coordination and harness-open detection on a limited-energy fixture.
4. Assemble cold storage	Discharge modules per Eaton procedure, verify voltage and polarity, fit local fuses/resistors, make the star connections and fit barriers. Connect matched-voltage branches only. Never connect a charged module directly across an empty bank.
5. Commission the direct path	Use the actual array or a characterized PV simulator. Demonstrate a 50 W load from PV with storage disabled, and measure the auxiliary demand.
6. Charge / test night service	Charge from solar to the ready state, isolate the solar input and record 16 hours at a metered 50 W load. Log all bank/module voltages, input/output energy and temperatures.
7. Repeat the daily cycle	Demonstrate at least three replenishment/no-solar cycles with no declining stored energy and acceptable handover transients. Exercise weak-source and prolonged-darkness shutdown.

Record output within 11.4-12.6 V, actual 50 W useful load, all auxiliary input paths, minimum module voltage, maximum temperature, leakage and conversion efficiency. A calculation is not a substitute for these records.

10 / Evidence boundary and references

Precise status: this is a completed functional system design and sizing study for a solar-powered, battery-free 50 W supply under the declared resource envelope. It is not a demonstrated generator, fabrication-ready custom control PCB or proof that the original RF/static mechanism works. Hardware-function and construction-release claims remain contingent on the specified review and tests.

The main topology and energy budget are internally consistent. The design uses manufacturer-backed component ranges. Integration work still requiring physical confirmation includes empty-bank current control, parallel charger behavior, loss limits, module leakage/conditioning, fail-safe supervision, pass-device SOA and fuse/resistor pulse coordination. Those uncertainties are not reported as passed tests.

S1 / REC Alpha Pure-R module datasheet, 420 W column
https://www.recgroup.com/sites/default/files/2025-01/Web_DS_REC_Alpha_Pure-R_EN.pdf

S2 / Eaton XLR-48 module datasheet, Technical Data 10510
https://www.eaton.com/content/dam/eaton/products/electronic-components/resources/data-sheet/eaton-xlr-48-supercapacitor-module-data-sheet.pdf

S3 / Eaton XLR module instruction manual, Technical Note 10509
https://www.eaton.com/content/dam/eaton/products/electronic-components/resources/technical/eaton-xlr-48-energy-storage-module-instruction-manual.pdf

S4 / ADI DC1924A demo manual and reference schematic
https://www.analog.com/media/en/technical-documentation/user-guides/dc1924afa.pdf

S5 / ADI LT8705 datasheet
https://www.analog.com/media/en/technical-documentation/data-sheets/8705ff.pdf

S6 / TI LM5069 datasheet, independent UVLO/OVLO arrangement
https://www.ti.com/lit/ds/symlink/lm5069.pdf

S7 / ST STPS10H100C diode datasheet
https://www.st.com/resource/en/datasheet/stps10h100c.pdf

S8 / Traco TEN 60WIR datasheet, 4812/7212 variants
https://www.tracopower.com/ten60wir-datasheet

S9 / Littelfuse KLKD series official datasheet
https://www.littelfuse.com/assetdocs/klkd-datasheet?assetguid=7b52eebb-c7cc-43df-a8da-9682a338e0f6

S10 / TE HSC2002R2J official product page
https://www.te.com/en/product-1-1630019-6.html

S11 / TI TLV1702 product and datasheet
https://www.ti.com/product/TLV1702

S12 / TI LM4040 reference datasheet
https://www.ti.com/lit/ds/symlink/lm4040.pdf

S13 / RECOM R-78HB5.0-0.5 official product page
https://recom-power.com/en/products/switching-regulators/switching-regulators-sip/rec-p-R-78HB5.0-0.5.html

S14 / TE EV200AAANA official product page
https://www.te.com/en/product-1618002-7.html

S15 / TI CSD19536KTT pass-MOSFET datasheet
https://www.ti.com/lit/ds/symlink/csd19536ktt.pdf

S16 / Eaton XLR product aid, 14.7 kg reference module mass
https://www.eaton.com/content/dam/eaton/products/electronic-components/resources/product-aid/eaton-xlr-supercapacitor-module-product-aid.pdf

Primary sources checked 26 October 2026. Manufacturer component specifications are distinct from the new assembly assumptions and calculated results in this design. Published Traco family data were accessible through indexed manufacturer documents; a direct PDF request returned access restrictions.